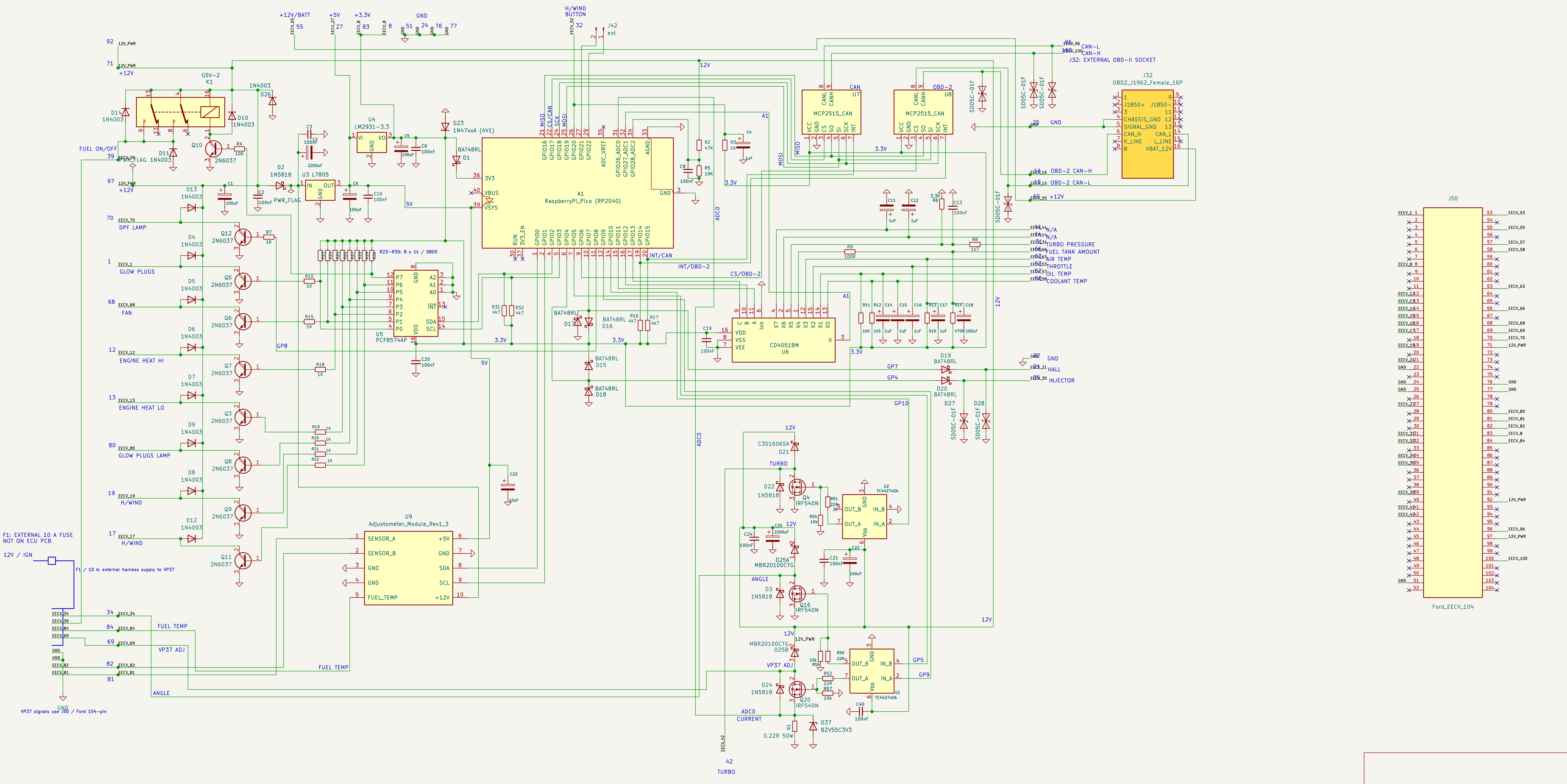

+12V/BATT
+5V
+3.3V
GND
H/WIND BUTTON
J42 ext
12V
K1 G5V-2
1N4003
D14
D10
D11
D26
Q10 2N6037
FUEL ON/OFF
PWR_FLAG
D2 1N5818
U3 L7805
U4 LM2931-3.3
D23 1N47xxA (4V1)
BAT48RL
D1
A1 RaspberryPi_Pico (RP2040)
3V3
VBUS
VSYS
GND
MISO
CS_CAN
SCK
MOSI
ADC_VREF
AGND
RUN
3V3_EN
U5 PCF8574AP
R23–R30: 8 x 1k / 0805
INT/CAN
INT/OBD-2
CS/OBD-2
U6 CD4051BM
U7 MCP2515_CAN
U8 MCP2515_CAN
CAN
OBD-2
J32 OBD2_J1962_Female_16P
J32: EXTERNAL OBD-II SOCKET
CAN-L
CAN-H
OBD-2 CAN-H
OBD-2 CAN-L
+12V
SD05C-01F
N/A
N/A
TURBO PRESSURE
FUEL TANK AMOUNT
AIR TEMP
THROTTLE
OIL TEMP
COOLANT TEMP
GND
HALL
INJECTOR
D19 BAT48RL
D20 BAT48RL
D27
D28
GP4
GP7
GP8
GP9
GP5
GP10
DPF LAMP
GLOW PLUGS
FAN
ENGINE HEAT HI
ENGINE HEAT LO
GLOW PLUGS LAMP
H/WIND
H/WIND
Q12 2N6037
Q5 2N6037
Q6 2N6037
Q7 2N6037
Q3 2N6037
Q8 2N6037
Q9 2N6037
Q11 2N6037
U9 Adjustometer_Module_Rev1_3
SENSOR_A
SENSOR_B
GND
GND
FUEL_TEMP
+5V
GND
SDA
SCL
+12V
F1: EXTERNAL 10 A FUSE NOT ON ECU PCB
12V / IGN
F1 / 10 A: external harness supply to VP37
FUEL TEMP
VP37 ADJ
FUEL TEMP
ANGLE
VP37 signals use J50 / Ford 104-pin
C3D16065A D21
TURBO
D22 1N5818
Q4 IRF540N
U2 TC4427xOA
MBR20100CTG D25A
ANGLE
D3 1N5818
Q16 IRF540N
MBR20100CTG D25B
12V_PWR
VP37 ADJ
D24 1N5818
Q20 IRF540N
U1 TC4427xOA
ADC0 CURRENT
R1 0.22R 50W
D37 BZV55C3V3
42 TURBO
ADC0
J50 Ford_EECV_104

PCB1: R / ceramic C = 0805; electrolytics = radial THT / 25V

CAN and R1: measured footprints; R1 mounting follows accepted mechanical baseline

TC4427: non-inverting, LOW = OFF; 12V_PWR supply.

Marcin Kielesiński PROFI-SOFT
Sheet: /
File: ECU.kicad_sch
Title: Fiesta VP37 ECU
Size: User
Date: 2026-11-09
Rev: v3.0-PCB1
KiCad E.D.A. 9.0.9
Id: 1/1